Multistage Sampling Design and Analysis for Detroit Transportation Study

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Introduction

This report outlines the sample design for a survey examining transportation security and use among Detroit residents. Transportation insecurity is a condition in which a person is unable to regularly get from place to place safely or on time due to a lack of resources necessary for transportation. It is associated with poor physical and mental health outcomes (McDonald-Lopez et al. 2023). In 2018, nearly one in four Americans experienced transportation insecurity (Murphy et al., 2022), and in Detroit, more than a third (36%) of residents experienced transportation insecurity in 2023 (Wileden et al., 2025).

Sample Design

Goals of the sample design

The primary objective of the sampling design is to obtain reliable estimates or several measures. These include of the proportion of residents living in a household without a vehicle (e.g., automobiles, vans, trucks), the proportion using public transportation as their main means of travel, the proportion who skip going somewhere because transportation problems, and the proportion who asked for transportation help in the past month. The survey will estimate these figures separately for four age groups of Detroit residents: 18–34, 35–49, 50–64, and 65+.

The sample design will include 1,000 respondents, with 250 in each age group, while accounting for differential response rates across domains.

Target population

The target population consists of all people 18 and older residing in the City of Detroit, Michigan. The sample aims for 250 completed responses in each age group to meet the desired precision and budget. Expected response rates for each group will help determine the required sample size. The required sample size for each domain is calculated by inflating the target number of completed responses with the expected response rate using this formula:

$$n_d^* = \frac{n_d}{RR_d}$$

n_d^* - required sample size for domain d

n_d – desired sample size in domain d (target number of completed responses)

RR_d – response rate for domain d

Each age group has different expected response rates, resulting in different required sample sizes for each domain. Table 1 shows the required sample size for each domain; the total across all domains is 2,154.

Table 1. Population and Sample Size by Age Domain

Domain	Population (N)	Expected Response Rate (RR)	Required sample size (n)	Sampling Rate (f)
18–34	25,000	0.35	714	0.0286
35–49	22,000	0.45	556	0.0253
50–64	20,000	0.5	500	0.025
65+	18,000	0.65	385	0.0214

Area sampling frame

The sampling frame was constructed using 2020 Decennial Census data at the tract and block group levels. Census tracts serve as primary sampling units (PSUs), block groups serve as secondary sampling units (SSUs), and individual respondents are the ultimate sampling elements. Tracts with zero adult population were removed, since they do not contribute to the target population. Tracts with fewer than two block groups were retained but flagged. In these tracts, the single block group is selected with certainty at the second stage. After these adjustments, the sampling frame includes 270 tracts and 612 block groups. The population size for the sampling frame by each domain is included in Table 1.

Sampling rates

Using the same sampling rate for all persons results in all persons having the same inclusion probability, which allows self-weighing, lower variance, and simpler analysis. Sampling rates are determined by the ratio of sample size to population size. When the same sampling rate is used for all persons, the resulting sample size is proportional to the population size. Sample size targets for each group cannot be achieved using this design unless they are proportional to the domain populations. Sampling rates for each age group were computed to determine this, using this formula:

$$f_d = \frac{n_d^*}{N_d}$$

The resulting sampling rates are shown in Table 1; they range from approximately 0.00425 to 0.00490 across age groups. Although these rates are relatively close, they are not identical. Since the rates are about the same, an approximately equal-rate design can be used. To control domain sample sizes and account for differential response rates, response-rate-adjusted measures of size can be used for allocation. This results in a design that is approximately self-weighting at the respondent level.

Assigning measures of size to units

A response-rate adjusted measure of size was used for both PSU and SSU selection. For each unit i, the MOS is defined as:

$$MOS_i = \frac{N_{i,18-34}}{0.35} + \frac{N_{i,35-49}}{0.45} + \frac{N_{i,50-64}}{0.50} + \frac{N_{i,65+}}{0.65}$$

This method gives greater weight to areas with more people in low-response groups. The sampling design then gives more selection probability to areas needing more sampling effort. This aligns selection probabilities with the effective sample size needs and improves the efficiency of the design.

Sample Selection

Stage 1: Selection of PSUs

Census tracts were chosen as PSUs through systematic sampling. Selection was done with probabilities proportional to size (PPS), using the response-rate adjusted MOS. The frame was sorted by tract identifier prior to selection, and a total of 40 PSUs were selected. The first-stage inclusion probability for tract i is:

$$\pi_{1i} = \frac{MOS_i}{\sum_j MOS_j} \times 40$$

Larger tracts, particularly those with higher concentrations of low-response populations, have a higher probability of selection.

Stage 2: Selection of SSUs

Within each selected tract, one block group was selected as the SSU using PPS systematic sampling with the same MOS definition. For tracts containing only one block group, that block group was selected with certainty.
The second-stage inclusion probability for block group j within tract i is:

$$\pi_{2j|i} = \frac{MOS_{ij}}{\sum_{k \in i} MOS_{ik}}$$

Appendix Table A1 lists the 40 chosen block groups and shows some characteristics like the percentage of people in each group. Most block groups have similar age distributions. One block group, tract 5202, block group 2, has 95% of its population in the 18–34 range. Other block groups have between 17.9% and 36.9% in this age range.

Stage 3: Selection of Persons

Within each selected block group, persons were sampled separately within each age domain. The number of persons selected from each domain was allocated proportionally to the domain population within sampled block groups, subject to the constraint that overall domain totals equal the required selected sample sizes.

The within-block-group sampling rate for domain d is:

$$f_{d,ij} = \frac{n_{d,ij}}{N_{d,ij}}$$

where $n_{d,ij}$ is the number of persons selected in age group d within block group j, tract i

Sampling effort is spread across areas in proportion to population. Workload per block group is about equal. The average number of selected persons per block group is about 54, but this may vary due to local population differences.

Selection of Persons from Households

Field staff will canvass each selected block group, listing all housing units in it. They will be provided with a map of the block group that shows the boundaries, a starting point and the direction they should travel as they list each housing unit. The listing will be used as a household sampling frame, from which a systematic sample of households will be selected. The total number of households that need to be selected are allocated across selected block groups in proportion to household counts adjusted by the inverse of the block-group selection probability, such that:

$$n_{ij}^{HH} \propto \frac{M_{ij}}{\pi_{ij}}$$

where n_{ij}^{HH} is the number of households selected in block group *j* within tract *i* and M_{ij} is the number of households in block group ij. Areas that were less likely to be selected in earlier stages receive more households with this allocation, helping to balance overall selection probabilities. The corresponding household selection probability is

$$\pi_{u|ij} = \frac{n_{ij}^{HH}}{M_{ij}}$$

Each selected household will provide roster information. Each eligible person will be assigned to a domain based on their age. Persons will be selected at different rates based on their domain. The rates are applied within households so that the total number of selected persons in each domain equals the domain target allocations across sampled block groups. The within-household probability for a person in domain d is:

$$\pi_{p|uij}(d) = \frac{f_d}{\pi_{1i} * \pi_{2j|i} * \pi_{u|ij}}$$

This conditional selection probability is specific to each domain and balances out unequal probabilities from the first two stages of sample selection so that the overall inclusion probability is constant and self-weighting is achieved for each domain.

Description of weighting plan

The overall inclusion probability for an individual in age group d, $\mathrm{household\ u}$, in block group j within tract i is:

$$\pi_{1i}\pi_{2j|i}\pi_{u|ij}\pi_{p|uij}(d)$$

The corresponding base weight is:

$$w_{ijup}(\mathrm{d}) = \frac{1}{\pi_{1i}\pi_{2j|i}\pi_{u|ij}\pi_{p|uij}(d)}$$

The design aims to achieve self-weighting within each age group, meaning that inclusion probabilities are approximately constant within domains. However, due to variation in age composition across block groups and differences in selection probabilities at each stage, exact self-weighting is not achieved at every stage.

Standard errors were obtained using design-based variance estimation methods that account for the multistage sampling design. The coefficient of variation (CV) of base weights within age groups ranges from approximately 0.39 to 0.42, indicating moderate dispersion. This level of variation is typical for multistage PPS designs.

Variance Estimation

Variance is estimated from the variability of PSU-level estimates across sampled tracts. To correctly estimate variances from a complex sample, the sample design structure, including strata, primary sampling units (PSUs), and survey weights must be considered. These elements determine how variability is measured, particularly through differences between PSUs within strata.

Strata are groups of PSUs, within which PSUs are independently selected. In this design tracts served as the PSUs and were ordered prior to systematic PPS sampling. This ordering provides implicit stratification but does not constitute explicit stratification so strata are not incorporated into the variance estimation. SSUs and other units such as block groups contribute to clustering as individuals within the same block group are more likely to have similar characteristics, including transportation behavior. This intra-cluster correlation increases variance relative to a simple random sample but these units are not treated as independent units in variance estimation.

Survey weights are incorporated into both point estimates and variance estimation. These weights are the inverse of the overall selection probability, including tract, block group, household, and person selection, as well as adjustments for nonresponse. Unequal weights contribute to additional variability and should be included in the variance estimation procedure.

Anticipated Precision

One of the variables of interest is the proportion of persons in Detroit who report public transportation as their main means of transportation. This proportion is calculated by:

$$\hat{p} = \frac{\sum_{(i,j.u,p)\in s} W_{ijup} y_{ijup}}{\sum_{(i,j.u,p)\in s} W_{ijup}}$$

where y_{ijup} = 1 if person k reports public transportation as their main mode, and w_{ijup} is the sampling weight.

Variance of this estimate can be computed using Taylor linearization which computes a linearized variable for each observation and then measures the variability of weighted PSU-level totals across strata, accounting for clustering and unequal weights:

$$\hat{V}(\hat{\theta}) = \frac{n_h}{n_h - 1} \sum_{i \in s_h} (\hat{z}_{hi} - \bar{z}_h)^2$$

The standard error (SE) is important in determining the variability of estimates. It is more easily interpretable because it uses the same unit as the estimate and can also be used to calculate confidence intervals.

$$SE(\hat{p}) = \sqrt{\widehat{Var}(\hat{p})}$$

The coefficient of variation (CV) measures the relative precision of estimated proportions and is helpful in comparing across domains.

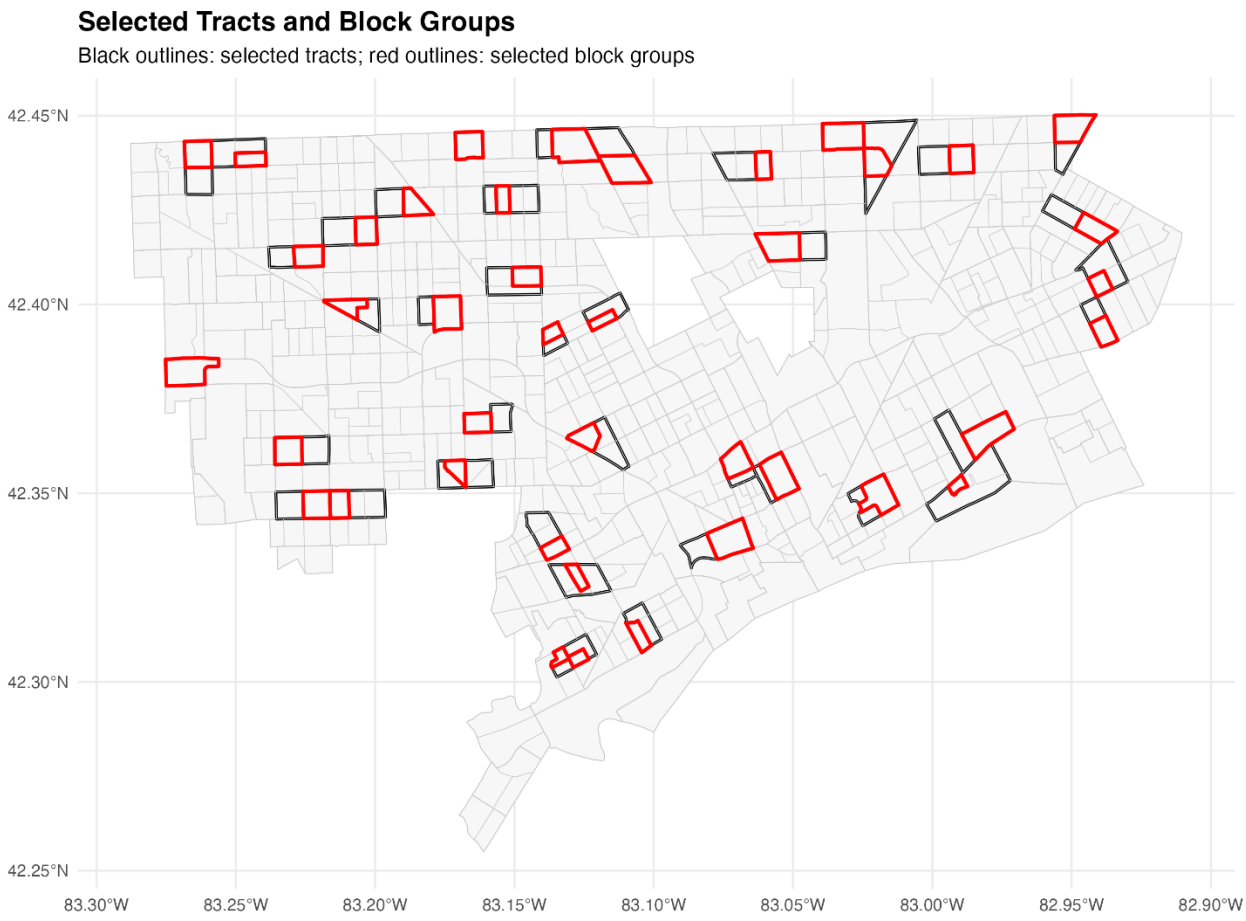
$$CV(\hat{p}) = \frac{SE(\hat{p})}{\hat{p}}$$

Summary

The survey uses a multistage probability sampling design to select persons in Detroit, with the objective of obtaining self-weighting samples within age domains and meeting domain-specific target sample sizes. The MOS used is based on response-rate–adjusted population counts that account for different expected response rates for each age group. The design proceeds in three stages. In the first two stages, census tracts and block groups are selected using PPS sampling based on the adjusted MOS. In the third stage, persons are selected through households using various methods. First, the required sample size for each domain is allocated across the selected block groups, in proportion to the domain's population count. Second, selected block groups are listed and these listings serve as a household sampling frame, from which a systematic sample of households will be selected. Lastly, persons are selected using domain-specific probabilities designed to achieve target sample sizes within each domain. This probability balances unequal selection probabilities from earlier stages.

Combining all stages, the overall inclusion probability for a person in domain d implies that all persons within a domain have the same probability of selection. This results in a design that is approximately self-weighting within domains and still meets domain-specific sample size targets.

Map



Codebooks

Github: https://github.com/Sagnik-Chakravarty/surv745-area-frame-multistage-survey-design/tree/main

Frame File Codebook

The frame files describe the area frame used for sampling. Two frame files are produced: one at the tract level and one at the block group level.

Tract Frame

Variable	Description
Tract	Census tract identifier used as the PSU ID.
NAME	Name or label of the census tract.
TotHH	Total number of households in the tract.
TotPerson	Total number of persons in the tract.
age18_34	Number of persons aged 18–34 in the tract.
age35_49	Number of persons aged 35–49 in the tract.
age50_64	Number of persons aged 50–64 in the tract.
age65p	Number of persons aged 65 and older in the tract.
age18p	Total number of persons aged 18 and older in the tract.
n_bg	Number of block groups contained in the tract.
zero_pop_flag	Indicator for whether the tract has zero adult population or zero eligible population.
lt2_bg_flag	Indicator for whether the tract has fewer than two block groups.
small_age18p_flag	Indicator for whether the tract has a small adult population, defined as fewer than 30 persons aged 18 and older.
MOS_adult	Measure of size based on the adult population (age18p).
MOS_rr_adj	Response-rate-adjusted measure of size based on age-specific population counts and expected response rates.
MOS	Final chosen measure of size used for sampling.

Block Group Frame

Variable	Description
Tract	Census tract identifier for the block group.
BlockGroup	Census block group identifier used as the SSU ID.
NAME	Name or label of the block group.
TotHH	Total number of households in the block group.
TotPerson	Total number of persons in the block group.
age18_34	Number of persons aged 18–34 in the block group.
age35_49	Number of persons aged 35–49 in the block group.
age50_64	Number of persons aged 50–64 in the block group.
age65p	Number of persons aged 65 and older in the block group.
age18p	Total number of persons aged 18 and older in the block group.
MOS_adult	Measure of size based on the adult population (age18p).
MOS_rr_adj	Response-rate-adjusted measure of size based on age-specific population counts and expected response rates.
MOS	Final chosen measure of size used for sampling.

Sample File Codebook

The sample file contains the selected sample units and the corresponding selection probabilities, within-domain sampling rates, and weights.

Variable	Description
Tract	Census tract identifier for the selected PSU.
BlockGroup	Census block group identifier for the selected SSU.
NAME	Name or label of the selected block group.
TotHH	Total number of households in the selected block group.
TotPerson	Total number of persons in the selected block group.

age18_34	Number of persons aged 18–34 in the selected block group.
age35_49	Number of persons aged 35–49 in the selected block group.
age50_64	Number of persons aged 50–64 in the selected block group.
age65p	Number of persons aged 65 and older in the selected block group.
age18p	Total number of persons aged 18 and older in the selected block group.
MOS	Final chosen measure of size used in selection.
pik1	First-stage inclusion probability for the selected tract.
pik2	Second-stage inclusion probability for the selected block group, conditional on tract selection.
n18_34	Number of selected persons allocated to the 18–34 age group in the selected block group.
n35_49	Number of selected persons allocated to the 35–49 age group in the selected block group.
n50_64	Number of selected persons allocated to the 50–64 age group in the selected block group.
n65p	Number of selected persons allocated to the 65+ age group in the selected block group.
f18_34	Within-block-group sampling rate for persons aged 18–34.
f35_49	Within-block-group sampling rate for persons aged 35–49.
f50_64	Within-block-group sampling rate for persons aged 50–64.
f65p	Within-block-group sampling rate for persons aged 65 and older.
pik_18_34	Overall inclusion probability for a person aged 18–34 in the selected block group.
pik_35_49	Overall inclusion probability for a person aged 35–49 in the selected block group.
pik_50_64	Overall inclusion probability for a person aged 50–64 in the selected block group.
pik_65p	Overall inclusion probability for a person aged 65 and older in the selected block group.

w_18_34	Base sampling weight for a person aged 18–34, equal to the inverse of the overall inclusion probability.
w_35_49	Base sampling weight for a person aged 35–49, equal to the inverse of the overall inclusion probability.
w_50_64	Base sampling weight for a person aged 50–64, equal to the inverse of the overall inclusion probability.
w_65p	Base sampling weight for a person aged 65 and older, equal to the inverse of the overall inclusion probability.
n_hh	Number of households selected in the block group for the household screening stage.
pi_hh_given_bg	Probability of selecting a household within the block group.
r18_34	Required within-household selection rate for persons aged 18–34 under the equal-probability feasibility design.
r35_49	Required within-household selection rate for persons aged 35–49 under the equal-probability feasibility design.
r50_64	Required within-household selection rate for persons aged 50–64 under the equal-probability feasibility design.
r65p	Required within-household selection rate for persons aged 65 and older under the equal-probability feasibility design.

Disclosure of AI Use

Portions of this project were developed with the assistance of generative AI tools (ChatGPT, OpenAI) to support code debugging, formatting, and clarification of methodological steps. The AI tool was used to refine R code structure, troubleshoot implementation issues, and improve the clarity and organization of written explanations.

All substantive decisions regarding the sampling design, including frame construction, measure of size specification, sample allocation, and interpretation of results, were made independently by the author. The final code, analysis, and written report were reviewed, verified, and edited to ensure accuracy and alignment with course requirements.

Table A1. Population Distribution and Selection Probabilities-Stage 1 and 2

Tract	Block Group	Population (N)	age18-34 (%)	age 35-49 (%)	age 50-64 (%)	age 65+ (%)	Selection Probability (Stage 1)	Selection Probability (Stage 2)
5001	2	2,665	112.8	74.5	76.7	35.3	0.2384	0.6626
5009	2	2,049	56.0	42.6	37.7	18.7	0.2201	0.5512
5014	2	990	36.1	28.1	28.7	14.7	0.2237	0.2671
5019	2	1,360	32.2	26.1	24.8	13.0	0.1460	0.5623
5033	2	1,600	34.2	27.0	21.8	13.4	0.1959	0.4743
5051	3	1,325	36.1	25.6	33.8	23.1	0.1850	0.4508
5061	1	1,875	19.0	12.0	12.6	8.1	0.1168	1.0000
5069	1	1,094	12.4	8.4	9.1	7.1	0.2122	0.2975
5080	1	1,098	27.3	21.2	27.3	8.6	0.0705	1.0000
5113	1	918	17.8	11.5	10.4	6.2	0.1292	0.4155
5138	1	1,205	20.7	12.3	17.0	11.4	0.1636	0.4980
5157	2	1,679	30.6	17.4	45.5	58.5	0.3365	0.3599
5167	3	1,254	20.0	16.8	22.1	22.0	0.2068	0.4150
5175	1	1,464	24.5	12.5	25.0	32.1	0.2014	0.4785
5202	2	2,615	224.3	4.2	3.6	1.4	0.3646	0.7422
5215	1	483	9.5	4.4	5.1	2.2	0.0887	0.3931
5232	3	1,111	10.7	10.0	5.9	2.4	0.1953	0.3277
5242	1	1,593	39.3	31.7	16.6	10.1	0.2511	0.3728
5257	4	1,368	17.3	18.1	11.1	5.1	0.2510	0.3033
5263	1	1,017	21.9	19.2	12.2	4.8	0.1837	0.3029
5304	2	729	11.0	5.5	6.4	5.0	0.0760	0.5581
5317	1	766	17.5	11.8	23.5	11.4	0.1014	0.4712
5334	1	615	13.9	9.2	9.5	6.1	0.1367	0.2879
5344	1	916	12.7	8.9	12.3	7.1	0.0934	0.6088
5356	1	659	10.6	9.1	12.1	10.5	0.2324	0.1812
5363	3	1,013	21.0	14.9	15.2	12.9	0.1680	0.3840
5371	1	1,134	19.7	14.2	12.7	10.8	0.1188	0.5904
5381	2	1,067	14.0	14.8	17.6	19.1	0.1412	0.4910
5386	3	1,111	13.1	9.9	13.2	10.3	0.2960	0.2405
5391	1	1,788	15.0	11.4	13.4	12.3	0.1099	1.0000
5396	1	1,368	68.1	46.4	60.2	42.7	0.1983	0.4351
5404	1	1,400	29.7	21.9	23.1	21.1	0.1645	0.5289
5409	1	993	13.3	10.3	11.7	11.4	0.2116	0.2828
5415	2	1,943	30.1	23.7	27.9	28.6	0.2321	0.5315
5423	1	858	15.9	11.8	10.9	10.2	0.1453	0.3790
5431	1	1,348	27.6	24.5	26.6	23.2	0.1516	0.5507
5440	1	2,534	95.2	64.9	77.9	34.3	0.1682	1.0000
5455	2	1,218	34.7	22.8	16.6	8.6	0.2234	0.3221
5459	2	1,897	27.0	17.1	14.6	5.6	0.2206	0.4971
5467	1	1,725	56.5	36.4	41.6	30.9	0.1749	0.6037

Table A2. Sampling Rates-Stage 3

Tract	Block Group	BG Person Sampling Rate (age18-34)	BG Person Sampling Rate (age 35-49)	BG Person Sampling Rate (age 50-64)	BG Person Sampling Rate (age 65+)
5001	2	0.0476	0.0611	0.0508	0.0553
5009	2	0.0487	0.0615	0.0522	0.0526
5014	2	0.0462	0.0595	0.0529	0.0515
5019	2	0.0491	0.0606	0.0518	0.0530
5033	2	0.0490	0.0621	0.0526	0.0526
5051	3	0.0484	0.0636	0.0517	0.0556
5061	1	0.0473	0.0627	0.0506	0.0553
5069	1	0.0471	0.0640	0.0481	0.0548
5080	1	0.0481	0.0619	0.0519	0.0588
5113	1	0.0496	0.0637	0.0493	0.0476
5138	1	0.0483	0.0612	0.0515	0.0546
5157	2	0.0489	0.0591	0.0514	0.0545
5167	3	0.0483	0.0619	0.0503	0.0539
5175	1	0.0463	0.0602	0.0514	0.0541
5202	2	0.0478	0.0638	0.0500	0.0667
5215	1	0.0449	0.0610	0.0526	0.0488
5232	3	0.0478	0.0630	0.0537	0.0492
5242	1	0.0488	0.0605	0.0495	0.0541
5257	4	0.0470	0.0609	0.0524	0.0568
5263	1	0.0498	0.0616	0.0522	0.0566
5304	2	0.0457	0.0612	0.0522	0.0556
5317	1	0.0497	0.0648	0.0509	0.0571
5334	1	0.0479	0.0631	0.0522	0.0548
5344	1	0.0469	0.0600	0.0483	0.0504
5356	1	0.0451	0.0614	0.0526	0.0530
5363	3	0.0469	0.0604	0.0486	0.0510
5371	1	0.0486	0.0625	0.0484	0.0570
5381	2	0.0471	0.0594	0.0498	0.0536
5386	3	0.0484	0.0642	0.0518	0.0561
5391	1	0.0486	0.0604	0.0513	0.0528
5396	1	0.0486	0.0625	0.0515	0.0534
5404	1	0.0485	0.0617	0.0506	0.0556
5409	1	0.0472	0.0610	0.0484	0.0552
5415	2	0.0474	0.0602	0.0513	0.0549
5423	1	0.0461	0.0621	0.0537	0.0571
5431	1	0.0463	0.0602	0.0517	0.0551
5440	1	0.0476	0.0613	0.0511	0.0520
5455	2	0.0464	0.0619	0.0485	0.0588
5459	2	0.0477	0.0601	0.0495	0.0556
5467	1	0.0485	0.0609	0.0502	0.0549